

Roll No.:

END SEMESTER EXAMINATION NOVEMBER 2025

Name of the Program: B.Tech Biotechnology

Semester: V

Name of the Course: Fundamentals of AI/ML

Course Code: TBT-505b

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)																																																					
(a)	Define Artificial Intelligence (AI) and Machine Learning (ML), and clearly distinguish between the two fields. Explain the concept of "Basic Notions of Learning" by describing how a system's performance improves with experience.					CO1 CO1																																																
(b)	i. Compare and contrast the three primary paradigms of Machine Learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. For each, state its goal, the type of data it requires, and provide a practical example.					CO1																																																
	ii. The following table shows data relating a student's study hours (x_i) to their test score (y_i).																																																					
	<table><tr><th>Observation (i)</th><th>Study Hours (x_i)</th><th>Test Score (y_i)</th></tr><tr><td>1</td><td>2</td><td>5</td></tr><tr><td>2</td><td>3</td><td>7</td></tr><tr><td>3</td><td>5</td><td>10</td></tr><tr><td>4</td><td>7</td><td>11</td></tr><tr><td>5</td><td>8</td><td>13</td></tr></table>					Observation (i)	Study Hours (x_i)	Test Score (y_i)	1	2	5	2	3	7	3	5	10	4	7	11	5	8	13																															
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Calculate Test Score (y_i) for Study Hours (x_i) = 4 with Linear Regression Model.																																																						
(c)	Explain the concept of Incremental Learning. Discuss a scenario where an incremental learning approach would be significantly more beneficial than batch learning for a real-world application, such as real-time threat detection or news feed personalization.																																																					
Q2	(20Marks)																																																					
(a)	This table shows data from 10 individuals regarding their Car Preference (Yes/No) based on vehicle and maintenance attributes.					CO2 CO2 CO2																																																
	<table><tr><th>ID</th><th>Vehicle Type</th><th>Age Group</th><th>Maintenance</th><th>Safety Rating</th><th>Preference (Target)</th></tr><tr><td>1</td><td>SUV</td><td>Young</td><td>High</td><td>Good</td><td>No</td></tr><tr><td>2</td><td>Sedan</td><td>Old</td><td>High</td><td>Good</td><td>Yes</td></tr><tr><td>3</td><td>SUV</td><td>Young</td><td>Low</td><td>Great</td><td>No</td></tr><tr><td>4</td><td>Sedan</td><td>Young</td><td>Low</td><td>Great</td><td>Yes</td></tr><tr><td>5</td><td>Hatchback</td><td>Old</td><td>Low</td><td>Good</td><td>Yes</td></tr><tr><td>6</td><td>Sedan</td><td>Young</td><td>High</td><td>Good</td><td>Yes</td></tr><tr><td>7</td><td>SUV</td><td>Old</td><td>Low</td><td>Great</td><td>No</td></tr></table>					ID	Vehicle Type	Age Group	Maintenance	Safety Rating	Preference (Target)	1	SUV	Young	High	Good	No	2	Sedan	Old	High	Good	Yes	3	SUV	Young	Low	Great	No	4	Sedan	Young	Low	Great	Yes	5	Hatchback	Old	Low	Good	Yes	6	Sedan	Young	High	Good	Yes	7	SUV	Old	Low	Great	No	
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	1. Calculate the Information Gain for the feature Vehicle Type and Maintenance. 2. Identify the Best Feature for the root node split. 3. Describe the resulting tree structure after the first split using the best feature. Specifically, identify which child node(s) become pure leaf nodes (i.e., where Entropy $H=0$).																											
(b)	Describe the comprehensive Data Mining Process, outlining its key sequential steps from data acquisition to final knowledge delivery. Explain the objective and expected output of each major stage.																											
(c)	Data Preprocessing is a critical stage in the data mining process. Explain the necessity of two key preprocessing tasks: Data Cleaning and Data Transformation. Provide an example of a technique used for each task.																											
Q3 (20Marks)																												
(a)	The following table shows data for 6 patients, including their weekly hours of moderate exercise (x) and whether they developed a specific health risk (Y). <table><tr><th>Patient ID</th><th>Exercise Hours (xi)</th><th>Health Risk (Yi)</th></tr><tr><td>1</td><td>1</td><td>1 (Yes)</td></tr><tr><td>2</td><td>2</td><td>1 (Yes)</td></tr><tr><td>3</td><td>4</td><td>1 (Yes)</td></tr><tr><td>4</td><td>5</td><td>0 (No)</td></tr><tr><td>5</td><td>7</td><td>0 (No)</td></tr><tr><td>6</td><td>8</td><td>0 (No)</td></tr></table> Assuming Exercise Hours (xi) = 4.5 calculate the Health Risk (Yi) with Logistic Regression.						Patient ID	Exercise Hours (xi)	Health Risk (Yi)	1	1	1 (Yes)	2	2	1 (Yes)	3	4	1 (Yes)	4	5	0 (No)	5	7	0 (No)	6	8	0 (No)	CO4 CO3 CO3
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(b)	Discuss the motivation for developing Artificial Neural Networks (ANNs) by describing the structure and key functional components of a Biological Neural Network (BNN). Explain how the concepts of dendrites, axon, and synapse are translated into the corresponding elements of an ANN (i.e., inputs, output, and weights).																											
(c)	Draw and label a typical architecture of a basic Feed-Forward Artificial Neural Network. Define the roles of the Input Layer, Hidden Layer(s), and Output Layer, and explain the function of the Activation Function within a neuron.																											
Q4 (20Marks)																												
(a)	Compare and contrast the K-Means Clustering algorithm with a Hierarchical Clustering algorithm (either Agglomerative or Divisive). Discuss the key differences in their approach to forming clusters, their computational costs, and their respective sensitivities to the initial state or number of clusters.						CO5 CO5 CO4																					
(b)	Explain the Maximum Likelihood Decomposition technique and its application to Normal Mixtures (Gaussian Mixture Models). How does this approach fundamentally differ from distance-based methods like K-Means in assigning data points to clusters, and what are the advantages of using a probabilistic model?																											
(c)	Define Deep Learning and its relationship to traditional Machine Learning. Introduce the concept of Deep Reinforcement Learning (Deep RL), explaining how it merges the capabilities of Deep Learning (e.g., using a neural network to approximate a																											

	function) with the principles of Reinforcement Learning to solve complex control problems.	
Q5	(20Marks)	
(a)	Define the concept of Uncertainty in AI and explain why it is a critical consideration for building robust intelligent systems. Introduce Bayesian Networks as a tool to model and manage uncertainty, specifying the two main components that define a Bayesian Network.	CO5 CO6
(b)	Explain the difference between Bayesian Network Learning and Bayesian Network Inference (Prediction). In the context of Decision Theory, describe how a Bayesian Network can be combined with a loss function or utility function to determine the optimal course of action under conditions of uncertainty.	CO6
(c)	Discuss two distinct Sample Applications of AI or Image Processing techniques specifically in the understanding and analysis of biological data (e.g., Genomics, Proteomics, Microscopy, or Medical Imaging). For each application, briefly describe the nature of the biological data being analyzed and the type of AI/ML technique employed.	

